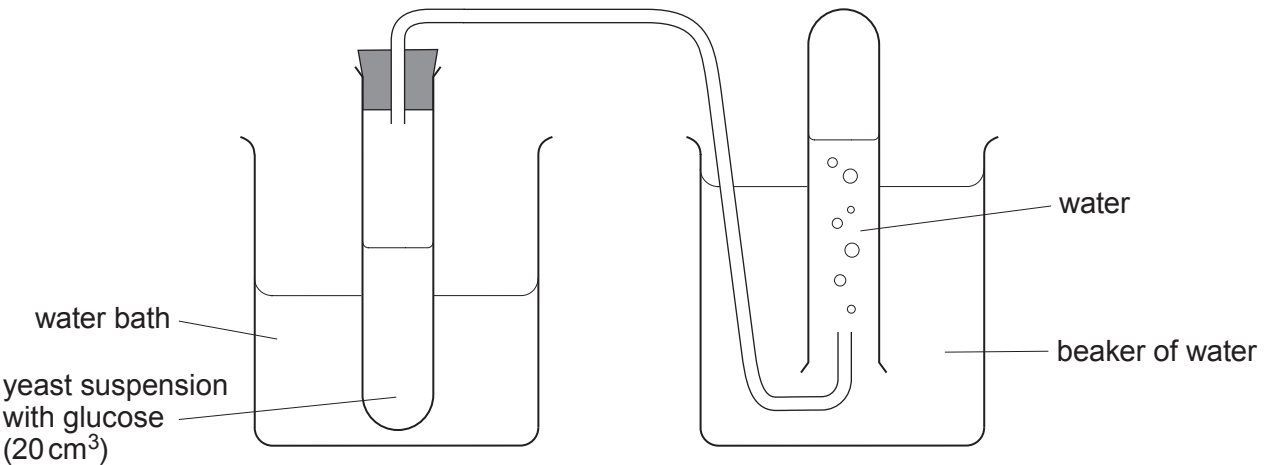


4. (a) A group of students carried out the following experiment to investigate the effect of temperature on the rate of aerobic respiration in yeast.

A yeast suspension containing glucose, made up in pH 7 buffer, was added to a boiling tube. The apparatus was set up as shown below.



The boiling tube was placed in a water bath at 10°C. The apparatus was left for three minutes. The students then counted the number of bubbles released every minute for five minutes.

The students repeated the procedure at 5°C intervals from 15°C to 40°C, each time using fresh yeast suspension with glucose from the same batch. Oxygen was available to the yeast throughout the experiment.

- (i) State the dependent variable in this experiment. [1]

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- (ii) Explain why the number of bubbles may decrease if a fresh yeast suspension with glucose was not used for each repeat. [1]

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- (iii) When investigating the rate of respiration, counting the number of bubbles is a limitation. Explain this limitation and suggest how it can be overcome. [2]

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- (iv) State the effect of respiration at 30°C on the pH of the yeast suspension in the absence of a buffer. Explain your answer. [2]

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- (v) The students then carried out the procedure with sucrose replacing the glucose in the yeast suspension.
The initial rate of respiration was found to be lower when sucrose was used as the substrate instead of glucose. Suggest why this may happen. [2]

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- (vi) ATP is a more useful immediate energy source for cell metabolism than either glucose or sucrose. Explain why ATP is described as an *universal energy currency*. [2]

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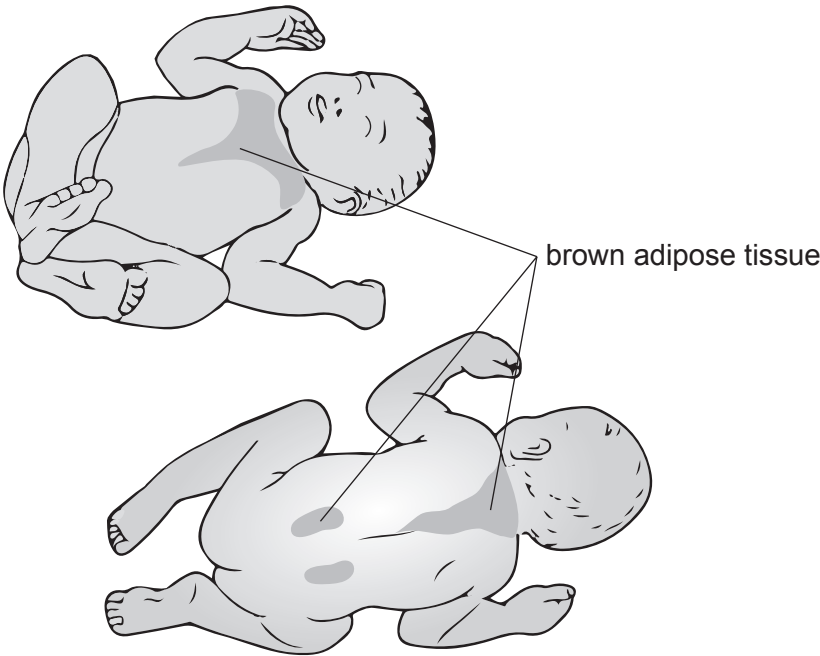
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(b) A group of scientists investigated the rate of respiration in brown adipose tissue. There is a high proportion of this tissue in newborn babies as shown below.



The cells in brown adipose tissue contain a much higher number of mitochondria than the cells of white adipose tissue; this gives the brown adipose tissue its colour.

The scientists isolated mitochondria from brown adipose tissue cells and used them to investigate some aspects of respiration. The mitochondria were placed in a solution with pyruvate as the respiratory substrate. An oxygen electrode was used to monitor the concentration of oxygen present in the solution.

(i) Explain why pyruvate was used as a respiratory substrate rather than glucose. [3]

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- (ii) When ADP was added to the solution, the concentration of oxygen present dropped by 40%. Explain this observation. [2]

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- (c) (i) State what happens to the energy that is not incorporated into ATP during respiration and the benefit of this to a newborn baby. [2]

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Adipose tissue samples can be fixed and cut into sections to be viewed through a microscope. The number of capillary lumens can be counted in a given area along with the number of fat cells (adipocytes). The results are expressed as number of capillaries per adipocyte.

The table below gives the capillary density in brown adipose tissue and white adipose tissue.

Adipose tissue type	Number of capillaries per adipocyte
white	0.4
brown	0.8

- (ii) Explain how the data in the table above shows that having a higher proportion of brown fat is of benefit to the baby. [3]

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